Mathematics Revision Test 35

Do you want to race against the clock?

1. Divide the sum of 53 and 65 by 2.

   _____

2. $\frac{3}{4} + \frac{3}{4} + \frac{1}{2} =$ _____

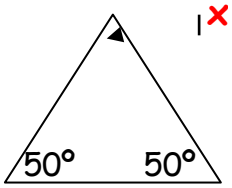
3. From the sum of 2 right angles take 33° _____

4. Put a decimal point in each of the numbers below so that the value of the 8 is **8 units.**

   a) 43**8**6 _____ b) 2**8**91 _____

5. Subtract 8 from the PRODUCT of 5 and 6 _____

6 $\frac{1}{20} = \frac{\square}{100}$



7. The known internal angles of this triangle are:

   Find × in degrees _____

8. 0.47 = _____ hundredths

9. Multiply the sum of 7 and 5 by 10 _____

10. Write $\frac{6}{1}$ an $\frac{8}{100}$ as a decimal fraction: _____

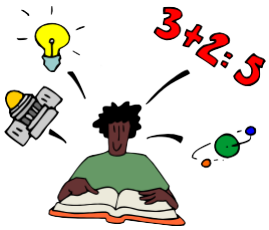
How many correct this week? _____

100% - ACE

Close to full marks - Almost ace

Make sure you always understand where you have gone wrong.

Time Taken: .....



Calculation Ladders

You must be so fast by now!

Three tests this week – read the notes on the left first.

A bit of extra help…

Now that you know how to find 10%, finding 20%, 30%, 40% etc is easy. For example:

30% is 3 lots of 10%.

Look at this:

Find 30% and 70% of £9.00

10% is $\frac{1}{1}$ of £9.00 = 90p

so 30% is 3 x 90p = £2.70
and 70% is 7 x 90p = £6.30

For 20%. 40%, 60% and 80% you can use another method:

20% = $\frac{20}{100} = \frac{2}{10} = \frac{1}{5}$

40% = $\frac{40}{100} = \frac{4}{10} = \frac{2}{5}$

60% = $\frac{60}{100} = \frac{6}{10} = \frac{3}{5}$

80% = $\frac{80}{100} = \frac{8}{10} = \frac{4}{5}$

For example, to find 60% of £15.00 you could use:

Method 1:

10% of £15.00 is £1.50
so 60% is 6 x £1.50 = £9.00

or Method 2:

60% is the same as $\frac{3}{5}$

$\frac{1}{5}$ of £15.00 is £3.00

$\frac{3}{}$

	Ladder 15	Ladder 16	Ladder 17
1.	fifty		
2.	find 20%		
3.	double it		
4.	+5		
5.	√n		—
6.	times by 9		
7.	double it		
8.	20% of this		
9.	find 50%		
10.	n^2		
	Your answer is		

so $\frac{}{5}$ is 3 x £3.00 = £9.00

Your answer is

Your answer is

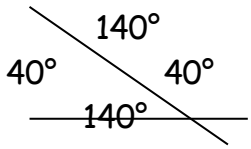
Are you STILL enjoying these?

Angles Handy Hints

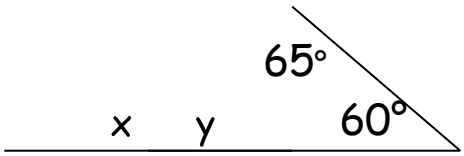
Read through last week's sheet on Angle Facts. Make sure you understand everything – they are straightforward.

Now read through the following carefully:

1. Opposite angles are equal:



2. There are 180° in a straight line. There are 180° in a triangle.
   If you remember these two easy rules you can easily solve the following problem:
   Find Angle x



First of all you need to know what the angle is at y.
The two angles you know in the triangle are 65° + 60° = 125°
So angle y is 180° - 125° = 55°
Therefore angle x is 180° - 55° = 125°
This method ensures that you understand what you are doing.

Simple method to calculate angle x: 65° + 60° = 125° This is angle x!

You are now ready to play the Angles Game. Use up all the cards. You may check your working out on a calculator if you wish!

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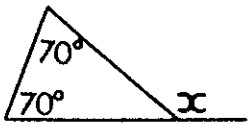
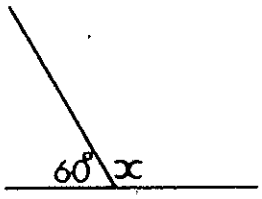
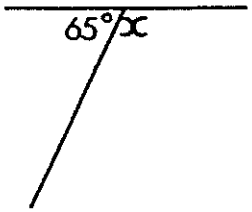
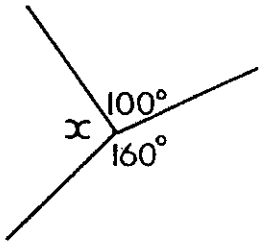
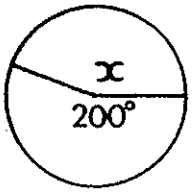
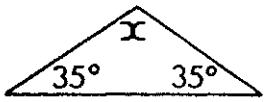
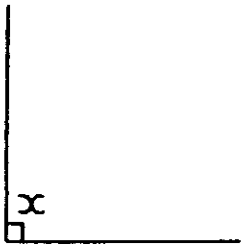
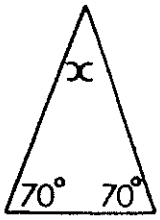
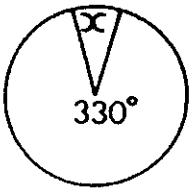
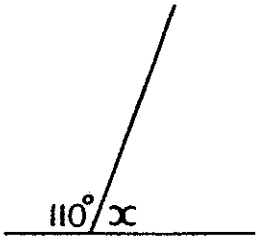
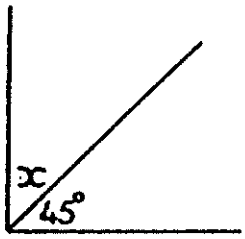
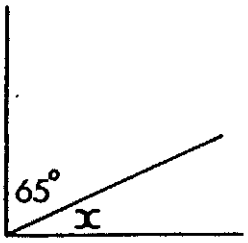
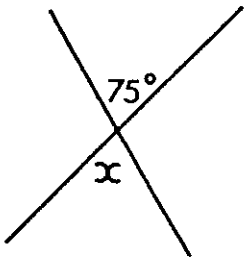
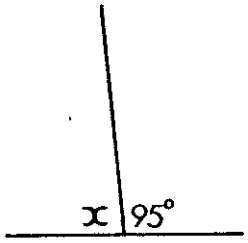
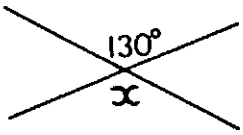
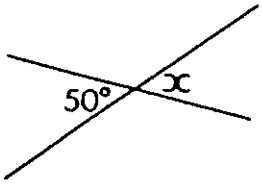
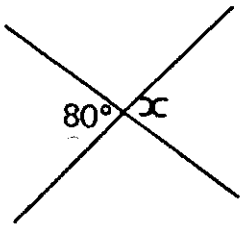
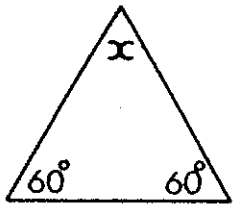
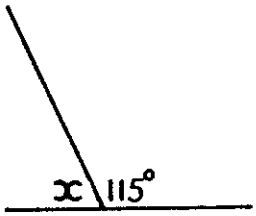
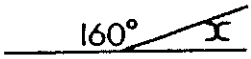
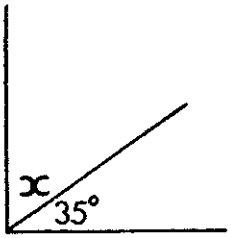
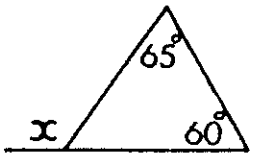
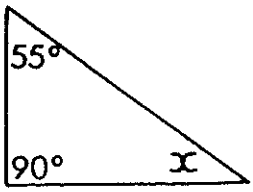
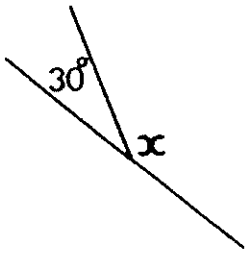
Angles Game

Cut out all the cards and match the 'x' angles shown to the correct value.

110°		50°	
160°		70°	
120°		75°	
80°		90°	
85°		40°	
20°		55°	

Angles Game

Cut out all the cards and match the 'x' angles shown to the correct value.

60°		35°	
115°		25°	
140°		45°	
30°		65°	
100°		130°	
150°		125°	

Glossary of Maths terms

Area	Amount of surface covered and measured in square units.
Co-ordinates	Ordered pair of numbers which position a point on a grid, e.g. (5,6).
Difference	An amount that separates two numbers, e.g. the difference between 9 and 20 is 11.
Even number	A whole number exactly divisible by two.
Factor	A whole number which divides exactly into a second number, e.g. 4 is a factor of 12.
Hexagons	A plane (2-D) shape with six sides. A regular hexagon has six equal sides and angles.
Mean/Average	A sum of numbers divided by the number of numbers counted, e.g. the mean of 3,5,7,and 9 is (3 + 5 + 7 + 9) ÷ 4 = 6
Median	Another measure of average where the numbers in a set are arranged in ascending order. The median is the middle number. If there are two middle numbers, the median is the mean of these two numbers.
Multiples	Multiples are found by multiplying by a specific number, e.g. multiples of 3 are 3,6,9,12,15, etc which are all in the three times table.
Odd numbers	Whole numbers which are not exactly divisible by two (leaving a remainder).
Parallelogram	A four-sided shape in which opposite sides are parallel and opposite angles are equal.
Perimeter	Distance or length all the way round the boundary of a plane shape.
Pie chart	A circular diagram used to represent data in which the parts or segments show different categories.
Product	A number obtained when two or more numbers are multiplied together, e.g. the product of 7 and 4 is 28.
Pyramid	A solid (3-D) shape which has triangular faces leaving a base to meet at a point.
Quadrilateral	A four-sided plane shape, e.g. square, rectangle and rhombus.
Rectangle	A four-sided shape in which opposite sides are equal and all the angles are right angles.
Rhombus	A parallelogram with all sides equal.
Sum	The result of adding two or more numbers e.g. the sum of 3, 6, and 9 = 3+6+9 = 18.
Symmetry: Reflective	Where one half of a plane shape corresponds exactly to the other half. Also known as mirror symmetry.
Symmetry: Rotational	A shape has rotational symmetry if when turned it still fits its outline exactly.
Triangle: Equilateral	A three-sided shape where all sides and angles are equal.
Triangle: Isosceles	A three-sided shape where two sides and two angles are equal.

1. Calculate (to the nearest whole number):

a/ 351 + 76 = _____ **b/** 9^2 = _____

c/ 1000 – 175 = _____ **d/** 1000 ÷ 9 = _____

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2. Calculate: 2100 - 686 _____

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3. Calculate 4/5 of 40. _____

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4. Peter got these marks in his school tests:

English .....21 out of 25

Science .... 8 out of 10

Maths .....38 out of 50

a/ What percentage did Peter get for Science? _____

b/ Write down the name of the test that Peter did best in. _____

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5. A recipe for 5 people uses 650g of potato. What weight of potato will be needed to make this recipe for 7 people? _____

---

6. This machine triples and then subtracts 7.

? → [ ] → 128

What number has been put in? _____

7. John parked in the car park from 10.15am to 2.35pm. The car park charges are £2.20 for the first hour and £1.50 for each additional half hour, or part of a half hour. How much did John pay? _____

8. A shop is giving a 20% discount on selected items in a sale. How much will a £975 plasma screen T.V. be reduced by if it is selected to be in the sale? _____

9. Louis will be z years old 5 years from now.
How old was he 3 years ago? _____

10. What is the next number in the series?

128 64 32 16 8 _____

11. **17 20 8 24 49**

Which of these numbers is:

a/ a factor of 100? _____

b/ a prime number? _____

c/ a multiple of 12? _____

12. The table shows the number of black and purple circles and rectangles in a box. What is the probability that a shape taken out of the box will be a purple circle? _____

	Black	**Purple**
Circles	**7**	**5**
Rectangles	**10**	**8**

13. A large packet of mints contains 300 mints. Work out the weight of the mints in one packet if each mint weighs 2.25 grams. _____

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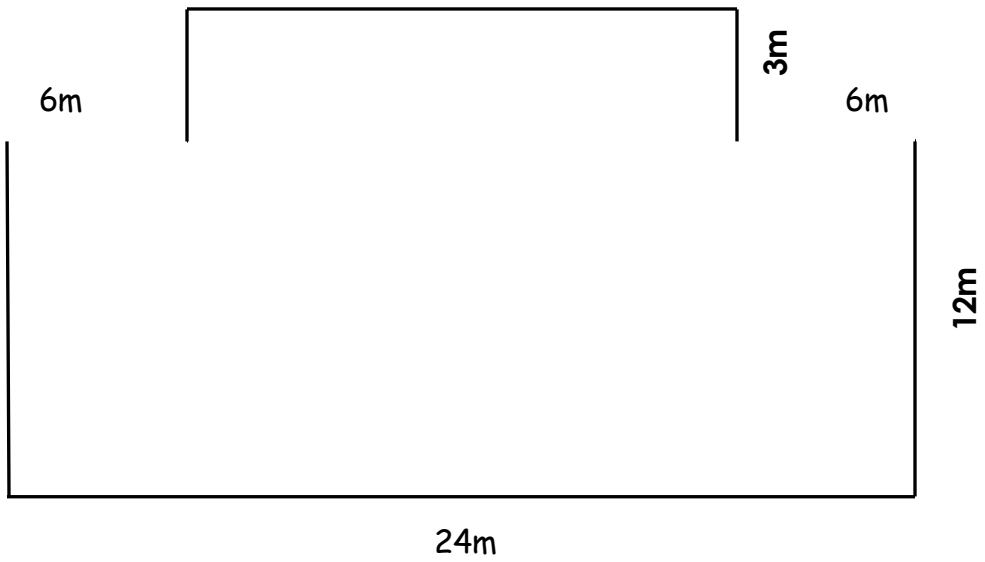
14. What area of turf is needed to cover the area of garden shown in the diagram? _____



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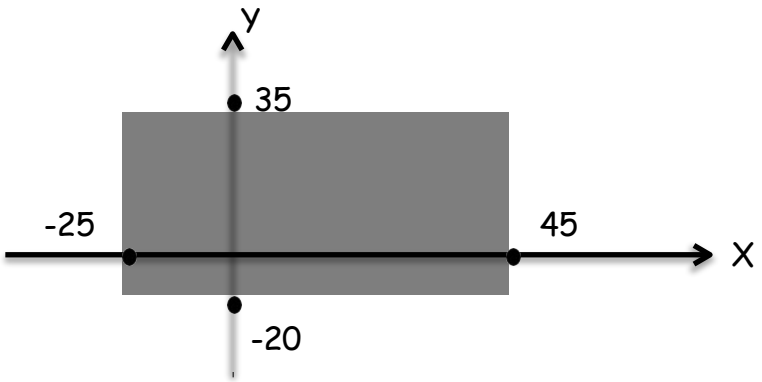
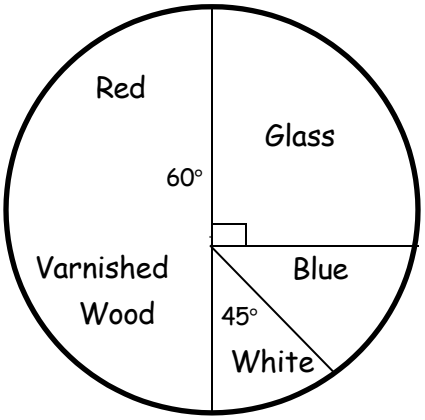


Which of these points are outside the shaded rectangle?

H. (12,27) **B. (33,-22)** **C. (-8,30)** **D. (10,-25)** **E. (-7,-1)**

_____

16. The pie chart shows the proportion of different back doors provided on a street of 120 houses.



a/ How many houses have varnished wood doors? _____

b/ What fraction (in lowest terms) of the houses on this street have back doors painted blue?

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17. Supermarket A sells crisps at £1.50 for a multipack of 6.
Supermarket B sells the same crisps at £2.20 for a multipack of 10.

Which supermarket is the more expensive and by how many pence per individual packet of crisps?

Supermarket _____
Difference per packet _____

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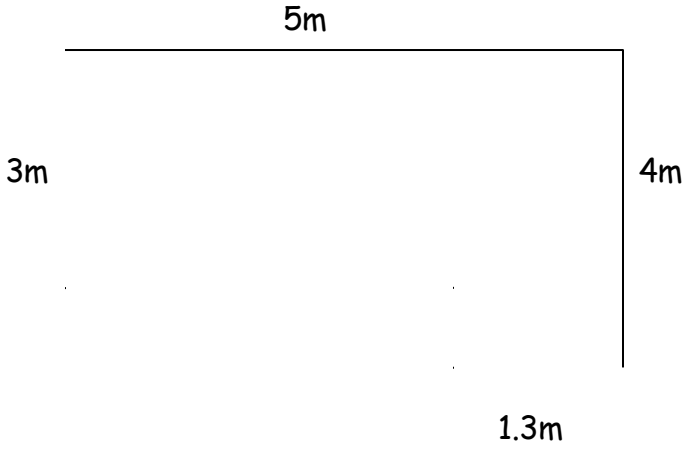
18. This is a plan of a patio.

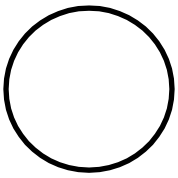


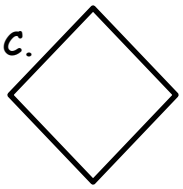
What area of stone paving will be needed to cover the patio?

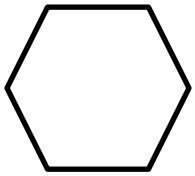
_____

19. Which shapes have exactly 4 lines of symmetry? _____

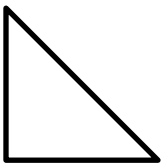
A.

B. 

C. 

D. 

E.

F. 

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20. Bus times between London and Crawley depend on the number of stops made on the way. Calculate the time taken by the **fastest** bus from the table below.

_____ hrs _____ mins

	Bus A	Bus B	Bus C	Bus D	Bus E
Departure Time	16.17	16.45	17.10	17.36	18.04
Arrival Time	17.22	17.55	18.06	18.45	19.02

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21. If 6x + 12 = 68 – x, what is x? _____

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22. Work out the value of:

17 – (8-5) x 4 _____

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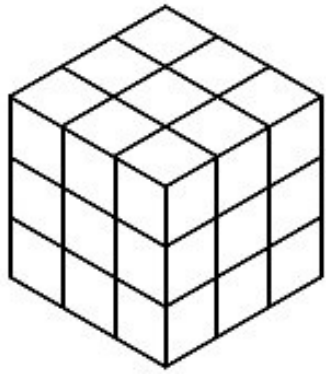
23. Peter likes to drink bottles of fizzy orange. The table shows the number of bottles Peter drank on each day of a particular week.

Day	Mon	Tues	Wed	Thurs	Fri	Sat	Sun
Number of Bottles	3	3	6	?	5	3	3

The mean is 4 and the median is 3. How many bottles did Peter drink on Thursday?

_____

24. Each of the cubes in this shape has sides of 2cm and weighs 20g.



a/ How many cubes are there in the bottom layer? _____

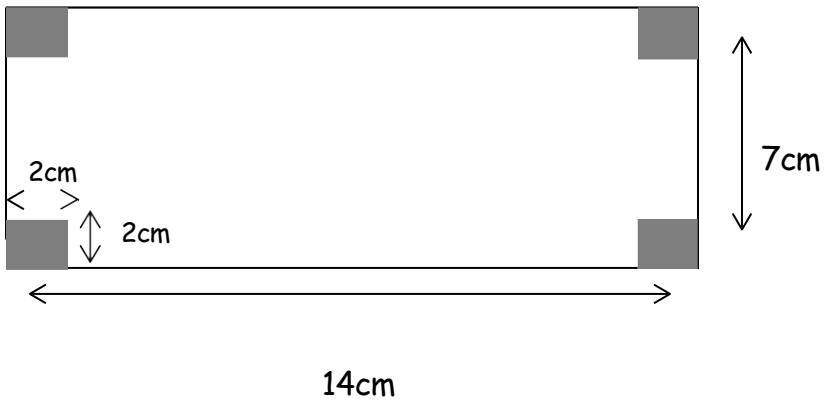
b/ What is the volume of this shape? _____

c/ How many cubes of the same size must be added to form another 2 layers underneath the bottom layer in the picture? _____

d/ How much will the enlarged 5-layer shape weigh? _____

---

25. The shaded corner squares are cut out of the 14cm x 7cm rectangle and the remaining shape folded to make a shallow open box.



What is the capacity of the box? _____

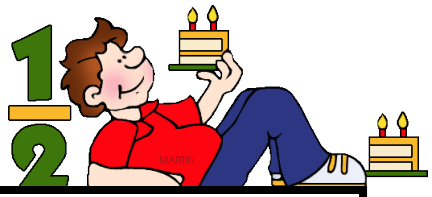
_____END OF TEST _____

If you have time

- Go back and complete any questions you missed out.

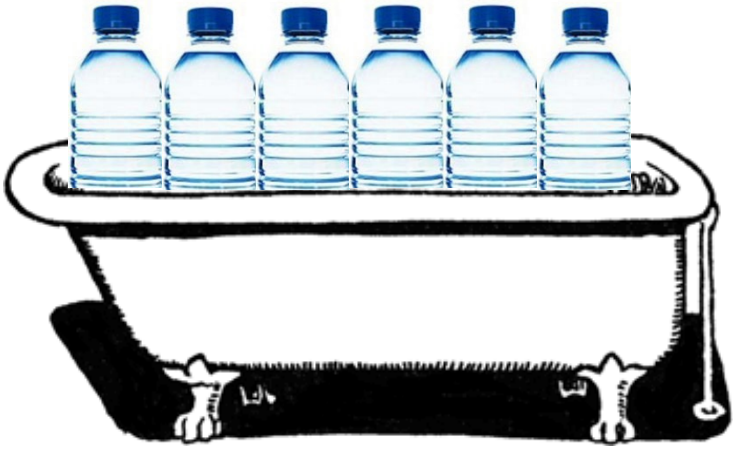
- Check your answers carefully.

Maths Reasoning



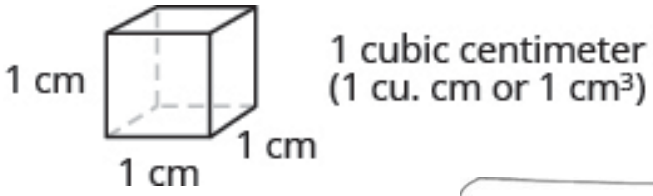
These questions are designed to get your mind whirring - so have a go. If you can't do it first time then try, try again. It is important to persevere.

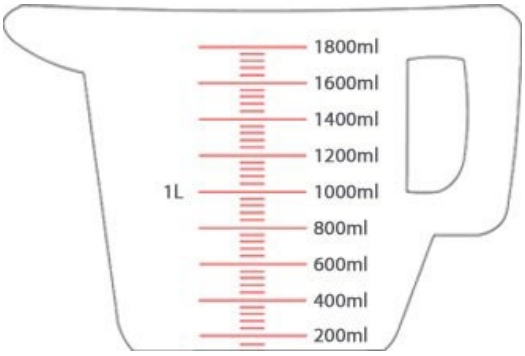
A bottle of water holds 50 centilitres.



If you wanted to fill a bath to the rim using only these bottles of water, estimate how many bottles would you need?

Hint: 1 cubic centimetre is the same as 1 millilitre.





ANSWERS - COMPLETE MATHS – PART 37

Maths

Revision Sheet

1. 59
2. 2
2. 147°
3. 438.6, 28.91
4. 22
5. 5
6. 80°
7. 47
8. 120
9. 0.68

Calculation Ladders

Ladder 1 = 81

Ladder 2 = £14

Ladder 3 = 75

Maths Type 2 Paper 1

Page 1

1. a. 427 b. 81
   c. 825 d. 111
2. 1414
3. 32
4. a. 80% b. English
5. 910g
6. 45

Page 2

7. £12.70
8. £195
9. z-8
10. 4
11. a. 20 b. 17 c. 24
12. 1 in 6

Page 3

13. 675g
14. 324m²
15. B and D

Page 4

16. a. 40 b. 1/8
17. Supermarket A, 3p
18. 16.3m²

Page 5

19. C and E
20. 56min
21. 8
22. 5
23. 5

Page 6

24. a. 9
    b. 216cm³
    c. 18
    d. 900g
25. 60cm³

Maths Reasoning

(The answer will vary based on the length, height and width that is measured or estimated, however the method of calculating shouldn't differ).

A length of 180cm, a width of 70cm and a height of 50cm for the bath is used in this example.

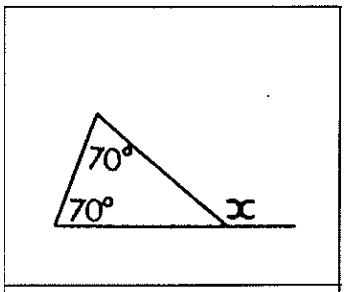
The volume of the bath is:

Volume = 180 x 70 x 50 = 630, 000cm³ or 630,000ml

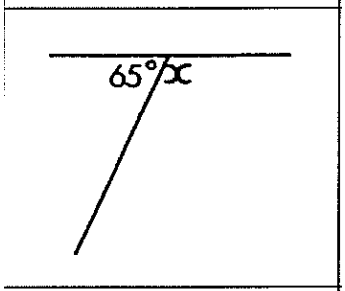
The number of bottles is found by dividing the total bath volume by the volume of a single bottle (be careful to make sure you are using the same units when doing this calculation):

$$No.\,of\ bottles = \frac{630{,}000\ ml}{500\ ml} = 1260\ bottles\ of\ water$$

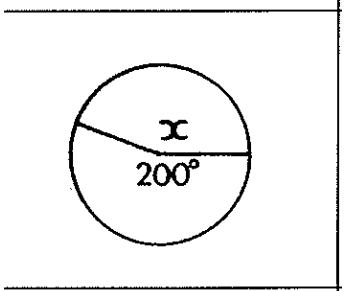
COMPLETE MATHS - ANGLES GAME ANSWERS - PART 37



x = 140°



x = 115°



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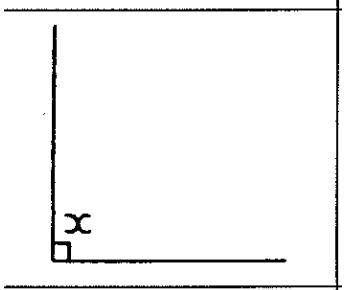
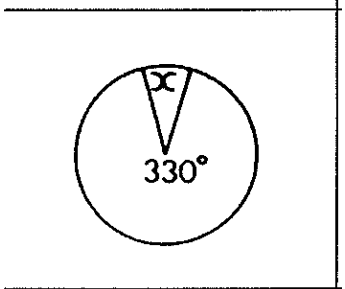
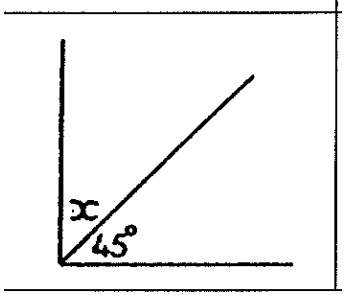


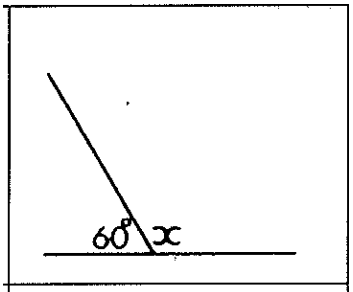
x = 90°



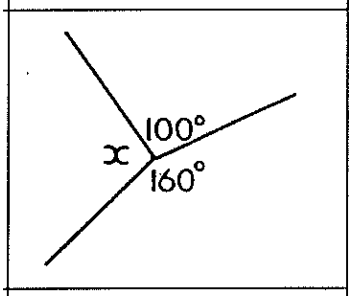
x = 30°



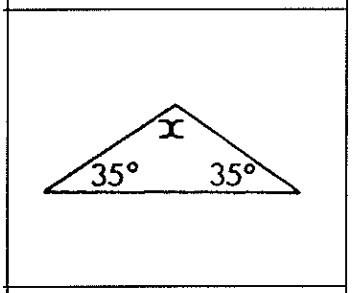
x = 45°



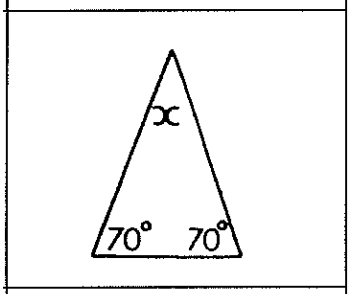
x = 120°



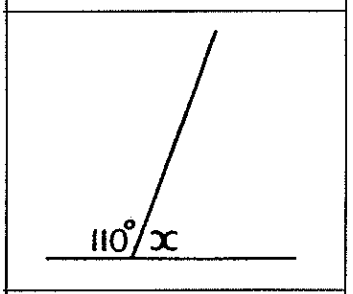
x = 100°



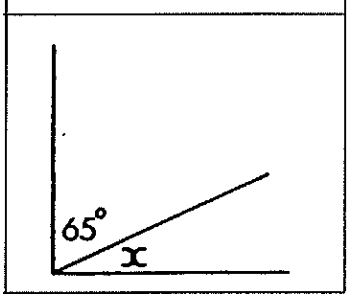
x = 110°



x = 40°

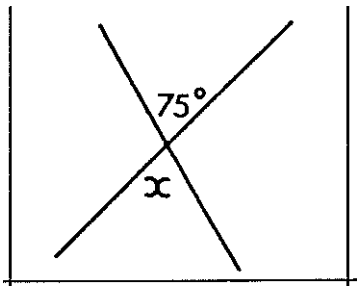


x = 70°

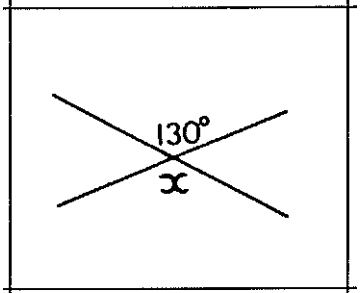


x = 115°

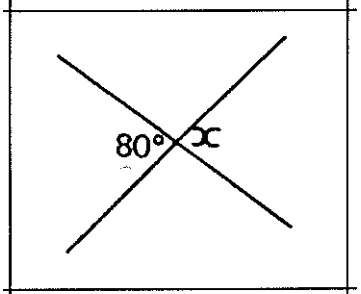
COMPLETE MATHS - ANGLES GAME ANSWERS - PART 37



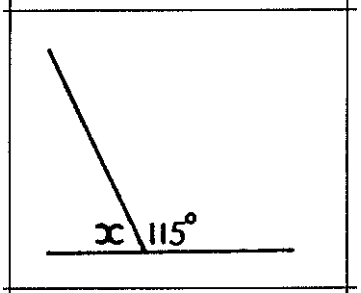
X = 75°



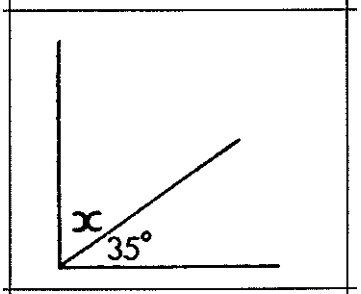
X = 130°



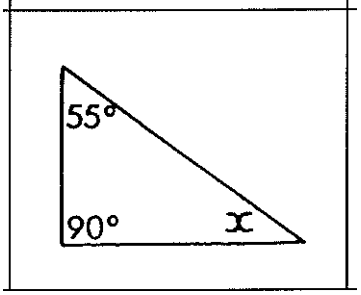
X = 80°



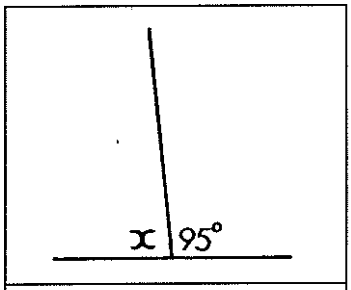
X = 65°



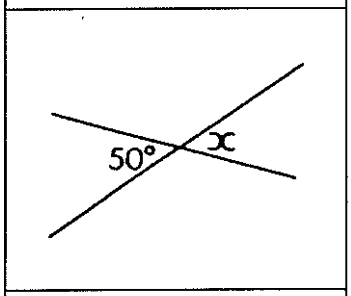
X = 55°



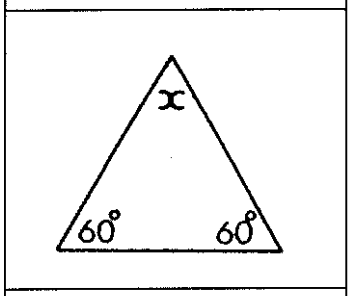
X = 35°



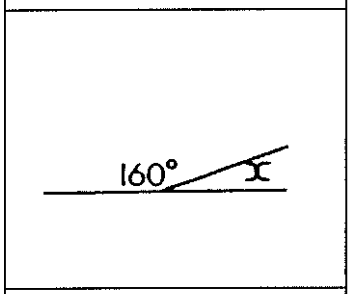
X = 85°



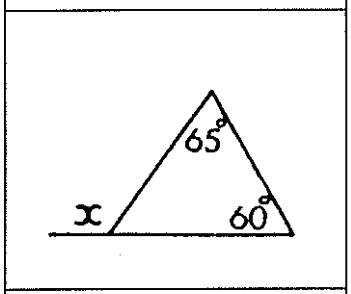
X = 50°



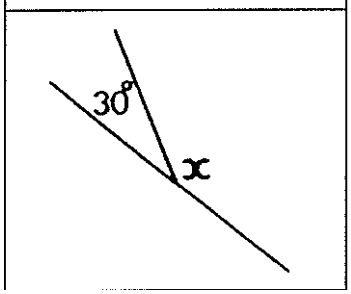
X = 60°



X = 20°



X = 125°



X = 150°